

BANK LOAN APPROVAL PREDICTION

AI/ML Internship Project

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Python | Google Colab | Pandas | NumPy | Scikit-Learn | Matplotlib | Seaborn

Project Objective

- Predict if a loan application should be approved or rejected.
- Automate the manual decision-making process.
- Analyze historical data to reduce human errors.
- Improve decision-making efficiency for financial institutions.

Dataset Overview

Information:

- **Source:** Loan Prediction Dataset
- **Size:** 614 Records | 13 Columns
- **Target:** Loan Status (Y/N)
- **Features:** Gender, Married, Education, Income, Loan Amount, Credit History, Property Area.

.. Train Shape: (614, 13)
Test Shape: (367, 12)

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
0	LP001002	Male	No	0	Graduate	No	5849	0.0	NaN	360.0	1.0	Urban	Y
1	LP001003	Male	Yes	1	Graduate	No	4583	1508.0	128.0	360.0	1.0	Rural	N
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0.0	66.0	360.0	1.0	Urban	Y
3	LP001006	Male	Yes	0	Not Graduate	No	2583	2358.0	120.0	360.0	1.0	Urban	Y
4	LP001008	Male	No	0	Graduate	No	6000	0.0	141.0	360.0	1.0	Urban	Y

Dataset Preview (train.head())

Project Workflow

- 1. Data Collection
- 2. Data Cleaning
- 3. Missing Value Handling
- 4. Exploratory Data Analysis
- 5. Feature Encoding
- 6. Train-Test Split
- 7. Model Training (Logistic Regression)
- 8. Prediction & Evaluation

Data Preprocessing

- Missing values identified in various columns.
- **Numerical:** Replaced using Median.
- **Categorical:** Replaced using Mode.
- Improved data quality for better performance.
- Categorical variables encoded into numerical values.

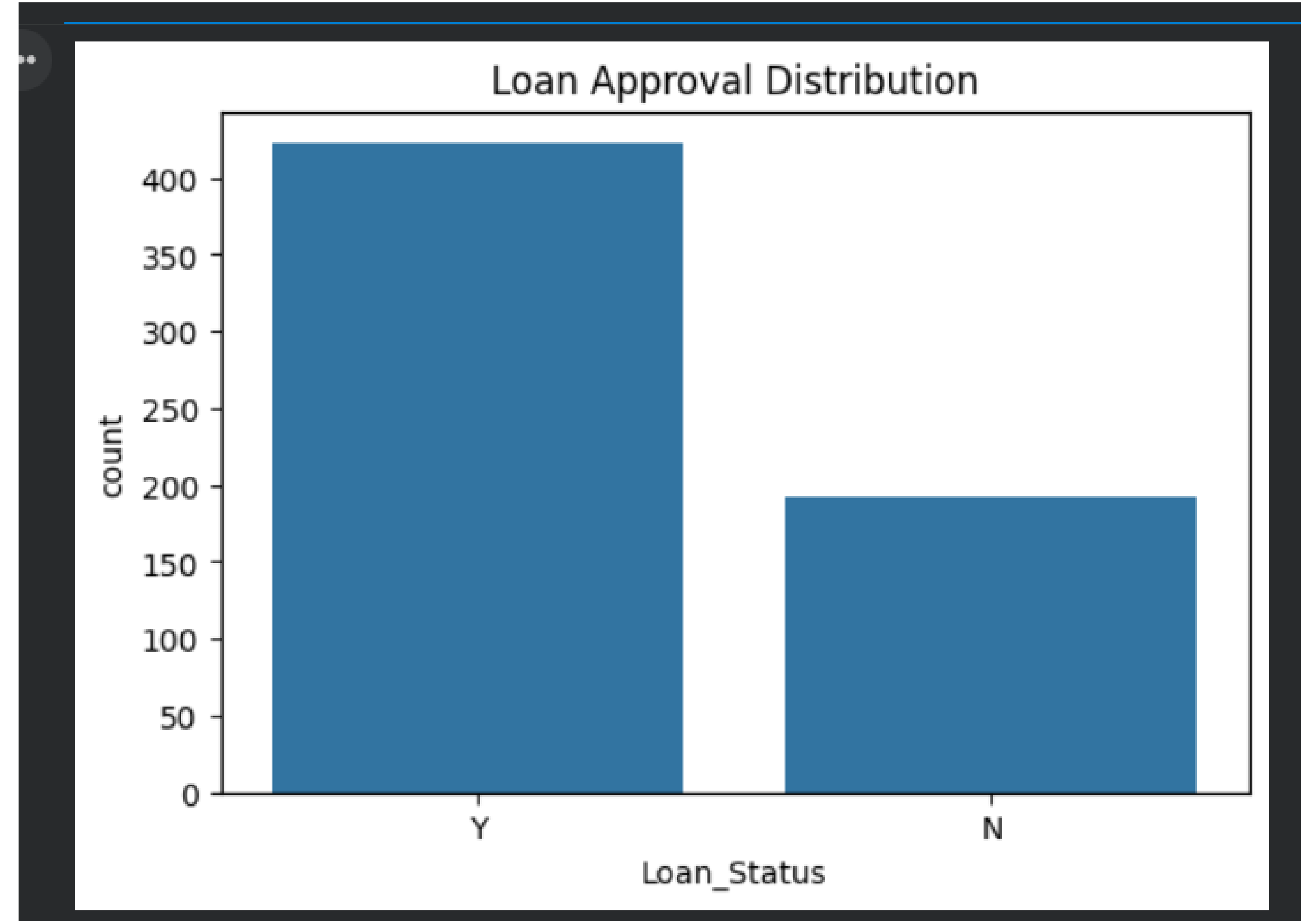
```
train.isnull().sum()
```

...	0
Loan_ID	0
Gender	13
Married	3
Dependents	15
Education	0
Self_Employed	32
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	22
Loan_Amount_Term	14
Credit_History	50
Property_Area	0
Loan_Status	0

dtype: int64

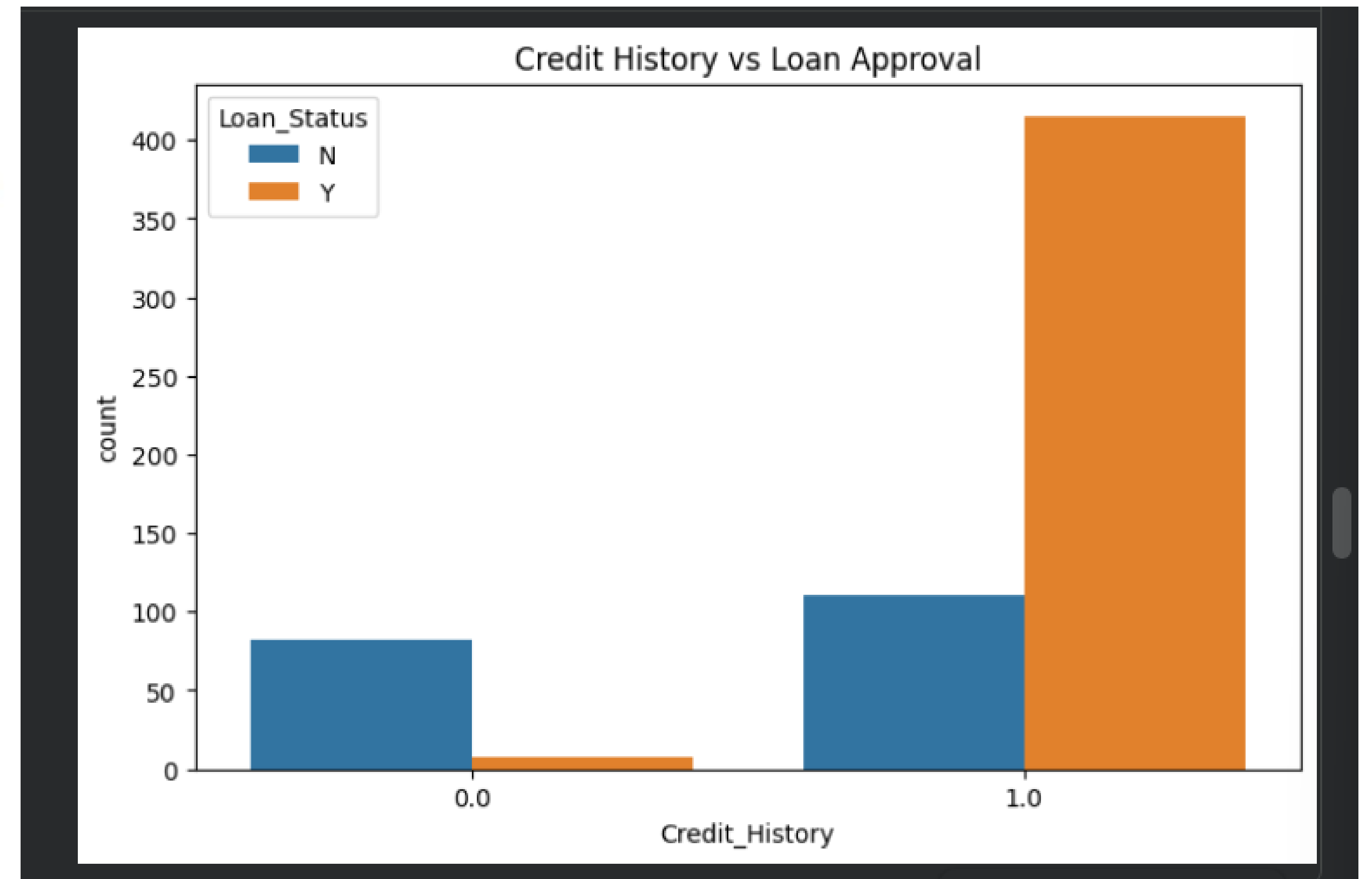
EDA: Loan Status Distribution

- Visualized the balance of approved vs rejected loans.
- **Observation:** Majority of the loan applications in the dataset were approved.



EDA: Credit History Analysis

- Impact of Credit History on loan approval status.
- **Observation:** Applicants with a good credit history (1.0) have a much higher chance of approval.



Model Results: Logistic Regression

- **Algorithm:** Logistic Regression.
- **Accuracy:** 78.86%
- Model performed well for binary classification.
- Confusion Matrix shows strong True Positive results.

```
accuracy = accuracy_score(  
    y_test,  
    y_pred  
)
```

```
print("Accuracy:", round(accuracy*100,2), "%")
```

```
Accuracy: 78.86 %
```


Final Prediction Output

- Tested model with sample applicant details.
- **Input:** Male, Married, Income 6000, Credit History 1.0.
- **Output:** LOAN APPROVED.

Model Accuracy: 84.55 %

Applicant Details:

Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	
0	1	1	0	1	0	6000	2000	150

✅ LOAN APPROVED

Conclusion & Future Scope

Conclusion:

- Successfully developed a loan prediction system.
- Credit History is the most important factor.
- Logistic Regression provided good baseline accuracy.

Future Scope:

- Use Random Forest or XGBoost for higher accuracy.
- Deploy as a real-time web application.